COLLABORATIVE PLATFORM : LEARN LINK

A MINI PROJECT REPORT

by

JOHNSON V MATHEW(VJC22CSD035)

in partial fulfillment for the award of the degree

of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND DESIGN

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

DEPARTMENT OF COMPUTER SCIENCE AND DESIGN

VISWAJYOTHI COLLEGE OF ENGINEERING AND TECHNOLOGY, VAZHAKULAM

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Mrs. JULITH JACOB

Assistant Professor, CSD Dept.

DEPARTMENT OF COMPUTER SCIENCE AND DESIGN

VISWAJYOTHI COLLEGE OF ENGINEERING AND TECHNOLOGY, VAZHAKULAM

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VISWAJYOTHI COLLEGE OF ENGINEERING AND TECHNOLOGY, VAZHAKULAM

Department of Computer Science and Design

Vision

Moulding socially dynamite Computer Science and Design engineers capable of driving technological advancements

Mission

1. To provide comprehensive education that combines technical knowledge, creativity and critical thinking through computer science principles and design methodologies.

2. Promote research and industry collaboration to empower our graduates for addressing real world challenges.

3. To cultivate ethical values, social responsibility and a commitment to lifelong learning among students.

Program Educational Objectives

Our Graduates

1. Shall have strong foundation in theoretical and practical aspects of Computer Science and Design principles to develop innovative solutions.

2. Shall possess effective communication, teamwork and leadership skills to collaborate with diverse stakeholders and to practice in a corporate firm.

3. Shall have potential to become entrepreneurs, innovators and pursue research in Computer Science and Design or related fields.

4. Shall have a strong sense of social awareness and stay updated with evolving design technologies.

Program Outcomes

1. **Engineering knowledge**:Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

2. **Problem analysis**:Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences

3. **Design / development of solutions**:Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety and the cultural, societal and environmental considerations.

4. **Conduct investigations of complex problems**:Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data and synthesis of the information to provide valid conclusions.

5. **Modern tool usage**:Create, select and apply appropriate techniques, resources and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

6. **The engineer and society:**Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

7. **Environment and sustainability:**Understand the impact of the professional engineering solutions in societal and environmental contexts and demonstrate the knowledge of and need for sustainable development.

8. **Ethics:**Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice

9. **Individual and team work:**Function effectively as an individual and as a member or leader in diverse teams and in multidisciplinary settings

10. **Communication:**Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations and give and receive clear instructions.

11. **Project management and finance**: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and unread in a team, to manage projects and in multidisciplinary environments.

12. **Life-long learning**:Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Specific Outcomes

1. Ability to integrate theory and practice to design software applications.
2. Able to apply computer science skills and design tools to analyse, design and model composite systems.
3. Ability to design and manage projects to develop a career in a related industry.

VISWAJYOTHI COLLEGE OF ENGINEERING AND TECHNOLOGY, VAZHAKULAM

DEPARTMENT OF COMPUTER SCIENCE AND DESIGN

BONAFIDE CERTIFICATE

Certified that the mini project work entitled " **COLLABORATIVE PLATFORM : LEARN LINK** " is a bonafide work done by **JOHNSON V MATHEW(VJC22CSD035)** in partial fulfillment of the award of the Degree of **Bachelor of Technology in Computer Science & Design** from APJ Abdul Kalam Technological University, Thiruvananthapuram, Kerala during the academic year 2024-2025

Internal Supervisor **External Supervisor**

Mini Project Coordinator **Head of Department**

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First and foremost, I thank **God Almighty** for His divine grace and blessings in making all these possible. May He continue to lead me in the years. It is privilege to render heartfelt thanks to most beloved Manager **Msgr. Dr. PIUS MALEKANDATHIL**, Director **Rev. Fr. PAUL PARATHAZHAM** and Principal **Dr. K K RAJAN** for providing me the opportunity to do this mini project during the sixth semester of B.Tech degree course. I am deeply thankful to the Head of the Department, **Mrs. SABITHA RAJU** for her support and encouragement. I would like to express my sincere gratitude and heartfelt thanks to my Mini Project Guide **Mrs. JULITH JACOB**, Assistant Professor, Department of Computer Science and Design for her motivation, assistance and help for the project. I also express sincere thanks to my Mini Project Coordinator **Mrs. ANJU PATHROSE,** Assistant Professor, Department of Computer Science and Engineering for her guidance and support. I also thank all the staff members of the Computer Science And Design Department for providing their assistance and support. Last, but not the least, i thank all my friends and family for their valuable feedback from time to time.

DECLARATION

I undersigned hereby declare that the mini project report **"COLLABORATIVE PLATFORM : LEARN LINK"**, submitted for partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology of the APJ Abdul Kalam Technological University is a bonafide work done by me under the supervision of **Mrs. JULITH JACOB**. This submission represents ideas in my own words and where ideas or words of others have been included, I have adequately and accurately cited and referenced the original sources. I also declare that I have adhered to the ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in my submission. I understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or formed the basis for the award of any degree, diploma or similar title of any other University.

Place : Vazhakulam

Date : March 31, 2025

JOHNSON V MATHEW

ABSTRACT

”Collaborative Platform : Learn Link” is an advanced collaborative platform that revolutionizes the way users engage in knowledge sharing, communication, and AI-driven decision-making. It is designed to integrate seamless interaction features and smart learning enhancements, providing an optimal environment for academic excellence and productivity. Set in a dynamic digital ecosystem, users take on the role of educators, students, and professionals striving to enhance collaboration. Learn-Link provides a structured approach to teamwork, brainstorming, and resource management, making it an essential tool for learning communities. The platform offers an engaging single-user and multi-user collaborative mode, allowing individuals and teams to optimize their learning strategies. Whether working independently or interacting in a group setting, ”Learn-Link” delivers a powerful, immersive experience that enhances learning through cutting-edge technology and innovative design.

Key Words :– React.js, Node.js, Gemini API

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List of Abbreviations

AI	Artificial Intelligence
LMS	Learning Management System
MFA	Multi-Factor Authentication
HTTPS	Hypertext Transfer Protocol Secure
SQL	Structured Query Language
API	Application Programming Interface
PDF	Portable Document Format
JSON	JavaScript Object Notation
RAM	Random Access Memory
HDD	Hard Disk Drive
SSD	Solid State Drive
VS CODE	Visual Studio Code

Chapter 1

INTRODUCTION

Learn-Link emerges as a transformative collaborative learning platform, designed to empower users by seamlessly integrating advanced technology with intuitive design. Built with a robust tech stack including React.js, Node.js, Express.js, and the Gemini API, Learn-Link delivers a dynamic user experience that fosters academic excellence. The platform enables users to engage in real-time collaboration, share educational resources, and leverage AI-powered tools to streamline their learning processes, all within an accessible and user-friendly interface.

Designed for students, educators, and learners, Learn-Link operates as a comprehensive solution, offering features like AI chat, audio calls, and co-editing capabilities for enhanced teamwork. By integrating with secure authentication systems like Auth0, the platform ensures user data protection while providing a seamless experience across devices. Learn-Link's intuitive design and AI-driven insights make it a powerful tool for organizing ideas, promoting knowledge sharing, and encouraging meaningful discussions.

What sets Learn-Link apart is its focus on accessibility and productivity, featuring tools like AI-generated summaries, quiz generation, and focus area extraction to support smarter decision-making. The platform's comprehensive offering includes real-time collaboration, content management, and communication features, all tailored to create an immersive learning environment.

Learn-Link promises users a rich and efficient learning experience that pushes the boundaries of collaborative education.

1.1 Problem Definition

Traditional learning platforms often suffer from several drawbacks compared to modern collaborative solutions. One of the key limitations is the lack of seamless collaboration. Many existing platforms fail to integrate real-time communication features like audio calls, creating a barrier to effective teamwork and discussion. This limited collaboration can make the learning experience feel isolated and less engaging. Another drawback is the absence of AI-driven tools for smarter decision-making. Without features like automated summaries or quiz generation, users struggle to efficiently process and retain information, leading to a less productive learning experience. Additionally, traditional platforms often have unintuitive designs, making it difficult for users to navigate and organize their ideas effectively.

In summary, the lack of seamless collaboration, absence of AI-driven tools, and unintuitive designs of traditional learning platforms can make them less appealing to users seeking more engaging and efficient learning experiences.

In response to these challenges, there is a pressing need for the development of innovative platforms that break away from traditional constraints. Developers are exploring new ways to enhance collaboration and productivity, such as through the use of real-time communication tools and AI-powered features, which can transform the learning experience by making it more interactive, efficient, and user-friendly.

1.2 Objective

The objective is to develop and implement an innovative collaborative learning platform, addressing the pressing challenges of traditional learning systems and user experiences. The primary goal is to create an interactive and efficient platform that leverages advanced technologies, specifically real-time collaboration tools and AI-powered features, to enhance the learning experience.

Firstly, the aim is to enhance collaboration through real-time communication and content sharing features. Learn-Link incorporates AI chat, audio calls, and instant messaging to facilitate discussions, brainstorming, and team learning. Features like annotations and co-editing further enhance teamwork, allowing users to work together seamlessly. Additionally, the platform enables users to upload and share educational materials, such as question papers and brainstorming artifacts, with automatic categorization and tagging to streamline content management. These capabilities ensure that users can organize their ideas efficiently and foster meaningful discussions, making the learning process more dynamic and engaging.

Secondly, the project seeks to improve the learning experience using AI-powered tools. Learn-Link integrates features like AI-generated summaries to create concise document overviews, quiz generation for self-assessment, and focus area extraction to identify key questions and categorize them. These tools provide users with smarter insights, enabling them to process information more effectively and focus on critical areas of study. By leveraging AI, Learn-Link enhances productivity and supports academic excellence, offering users a more intelligent and efficient learning experience.

The final objective is to ensure a seamless and secure user experience through a robust tech stack and authentication system. Learn-Link is built using React.js for a dynamic frontend, Node.js and Express.js for a scalable backend, and the Gemini API for advanced functionality. Additionally, Auth0 is integrated to provide secure user authentication and authorization, ensuring data protection and a smooth experience across devices. This comprehensive tech stack ensures that Learn-Link is accessible, reliable, and capable of meeting the diverse needs of its users.

1.3 Scope

The scope of collaborative learning platforms like Learn-Link is vast and ever-expanding, offering a range of possibilities that address many of the drawbacks of traditional learning systems. Such platforms have the potential to provide unparalleled levels of collaboration, enabling users to engage in real-time discussions and share resources seamlessly. This level of interactivity can help overcome the limitations of traditional platforms, such as the lack of teamwork and engagement, by providing a more connected and dynamic experience. Furthermore, platforms like Learn-Link can offer a greater sense of productivity and efficiency, allowing users to organize their ideas, streamline their learning processes, and access AI-driven insights for smarter decision-making.

Moreover, the adaptability of collaborative platforms can be significantly enhanced through the use of modern technologies and intuitive design, creating solutions that cater to diverse user needs. This can help address the issue of unintuitive designs in traditional platforms, as users can benefit from a more accessible and user-friendly interface. In the context of collaborative learning, Learn-Link offers a comprehensive solution that fosters academic excellence through enhanced collaboration and resource sharing. Meticulously crafted using React.js, Node.js, and Express.js, the platform integrates real-time collaboration, content management, and AI-powered tools to create an immersive learning environment. Learn-Link seamlessly blends productivity, accessibility, and innovation, promising users the ultimate collaborative learning experience.

Chapter 2

LITERATURE SURVEY

Collaborative learning platforms have evolved significantly in recent years, with many new solutions offering various features for knowledge sharing and communication. The demand for such platforms is expected to grow as they provide more efficient and engaging learning experiences compared to traditional methods.

2.1 Evaluating Experiences in Different Collaborative Learning Platforms [1]

The authors evaluate three scenarios across two types of collaborative platforms: traditional learning management systems (LMS) and modern AI-enhanced platforms. The scenarios include content sharing, real-time collaboration, and AI-assisted learning. The study compares engagement, knowledge retention, user satisfaction, and overall experience. Initial findings suggest that AI-enhanced platforms better support scenarios involving real-time interaction and personalized learning. The AI-assisted learning scenario showed higher engagement but required more technical infrastructure, while the collaborative tasks were more engaging in terms of productivity.

2.1.1 Advantages

- Engagement: Both traditional LMS and modern platforms offer engagement features, but AI-enhanced platforms provide more personalized experiences.
- Productivity: Collaborative tasks were more productive and efficient on modern platforms, especially those with real-time co-editing features.
- Satisfaction: Users reported higher satisfaction with AI-powered features like document summarization and quiz generation.
- Performance Metrics: Metrics such as knowledge retention, task completion time, and user satisfaction were rated higher on modern platforms.

2.1.2 Disadvantages

- Technical Requirements: Modern platforms require more robust technical infrastructure, which can be a barrier for some users.
- Learning Curve: Some users reported a steeper learning curve when adapting to AI-powered features.
- Privacy Concerns: Data collection for personalization raised privacy concerns among some participants.

2.2 Learning with AI: How Artificial Intelligence Affects User Satisfaction [2]

The authors convey that AI-enhanced learning platforms, such as those incorporating natural language processing and machine learning, offer a more personalized and efficient learning experience compared to traditional platforms. Their research explores the impact of AI tools on user satisfaction by having participants use both AI-enhanced and standard platforms. They measured satisfaction using a standardized scale, which included constructs like usability, efficiency, and engagement. The study found that AI tools enhanced overall satisfaction, engagement, and learning outcomes compared to traditional setups.

2.2.1 Advantages

- Higher Satisfaction: Users rated higher satisfaction when using AI-powered features like automated summaries and quiz generation.
- Efficiency: Participants completed tasks more efficiently with AI assistance, indicating time-saving benefits.
- Personalization: AI-driven recommendations and insights were highly valued for tailored learning experiences.

2.2.2 Disadvantages

- Limited Evaluation: The study focused on specific AI tools, and results may not generalize to all AI-enhanced platforms.
- Dependency: Some users became overly reliant on AI features, potentially hindering critical thinking skills.
- Technical Issues: Occasional bugs in AI features were reported, affecting user experience.

2.3 Real-Time Collaboration in Learning: User Experience and Performance [3]

The authors suggest that real-time collaboration tools have great potential for enhancing learning but face technical and usability challenges. They conducted an experiment comparing traditional asynchronous collaboration with real-time co-editing and communication features. They used performance metrics, questionnaires, and subjective reports to evaluate the experience. Despite a slight learning curve, users overwhelmingly preferred the real-time collaboration features. The study indicates that users adapted quickly to the real-time context, highlighting the potential of these tools for collaborative learning.

2.3.1 Advantages

- Strong User Preference for Real-Time Features: Most users preferred real-time collaboration, reporting higher levels of engagement and teamwork.
- Faster Task Completion: Groups using real-time features completed collaborative tasks more quickly.
- Improved Communication: Instant messaging and co-editing features facilitated clearer and more efficient communication.

2.3.2 Disadvantages

- Technical Challenges: Some users experienced latency issues during peak usage times.
- Distraction Potential: Real-time notifications were occasionally distracting for some participants.
- Limited Offline Functionality: The reliance on real-time features reduced functionality in offline scenarios.

The literature surveys highlight the transformative potential of modern collaborative learning platforms like Learn-Link, emphasizing their ability to offer users efficient, personalized, and engaging learning experiences. Users consistently report higher satisfaction and productivity with features like AI-powered tools and real-time collaboration. However, challenges such as technical requirements and privacy concerns remain, indicating areas for further refinement. Overall, the surveys underscore the growing significance of integrated, intelligent learning platforms in education and professional development, suggesting a future where such tools are central to knowledge sharing and collaboration.

Chapter 3

PROPOSED SYSTEM

The proposed system for developing ”Learn-Link” involves a comprehensive approach that uses React.js for frontend development, Node.js with Express.js for backend services, and AI integration via the Gemini API for enhanced learning tools.

React.js is a powerful JavaScript library used for building dynamic user interfaces. It supports component-based architecture, enabling efficient UI updates, desktop-optimized design, and seamless user interactions. React.js excels in creating robust web applications with reusable components, making it ideal for Learn-Link’s desktop website interface. Here, we utilize React.js to craft an intuitive and responsive frontend, featuring real-time collaboration tools like live chat, co-editing panels, and content management dashboards. Its flexibility ensures a smooth experience on desktop browsers, enhancing usability for learners.

Node.js, paired with Express.js, forms a robust JavaScript runtime and web framework for server-side development. Node.js enables scalable, event-driven applications, while Express.js simplifies API creation, routing, and middleware integration. This duo powers Learn-Link’s backend, handling user authentication, real-time communication, and data management. The system leverages Node.js for efficient handling of desktop user requests and Express.js to streamline API endpoints for features like quiz generation and document summarization. Additionally, the Gemini API integrates AI capabilities, providing summaries, focus area extraction, and quiz creation, enriching the website’s educational tools. Authentication is secured via Auth0, ensuring safe user access and data protection.

In Learn-Link, real-time collaboration is facilitated through WebSocket protocols within Node.js, enabling live chat, video calls, and instant messaging for desktop users. The website employs a distributed database system for high availability, using SQL for structured data queries. This ensures seamless content sharing, such as educational materials and brainstorming artifacts, with automatic categorization and tagging. The AI-driven tools enhance learning by analyzing uploaded content and generating personalized outputs, while HTTPS encryption and multi-factor authentication (MFA) safeguard user data, fostering a secure and interactive desktop learning environment.

3.1 Architecture Diagram

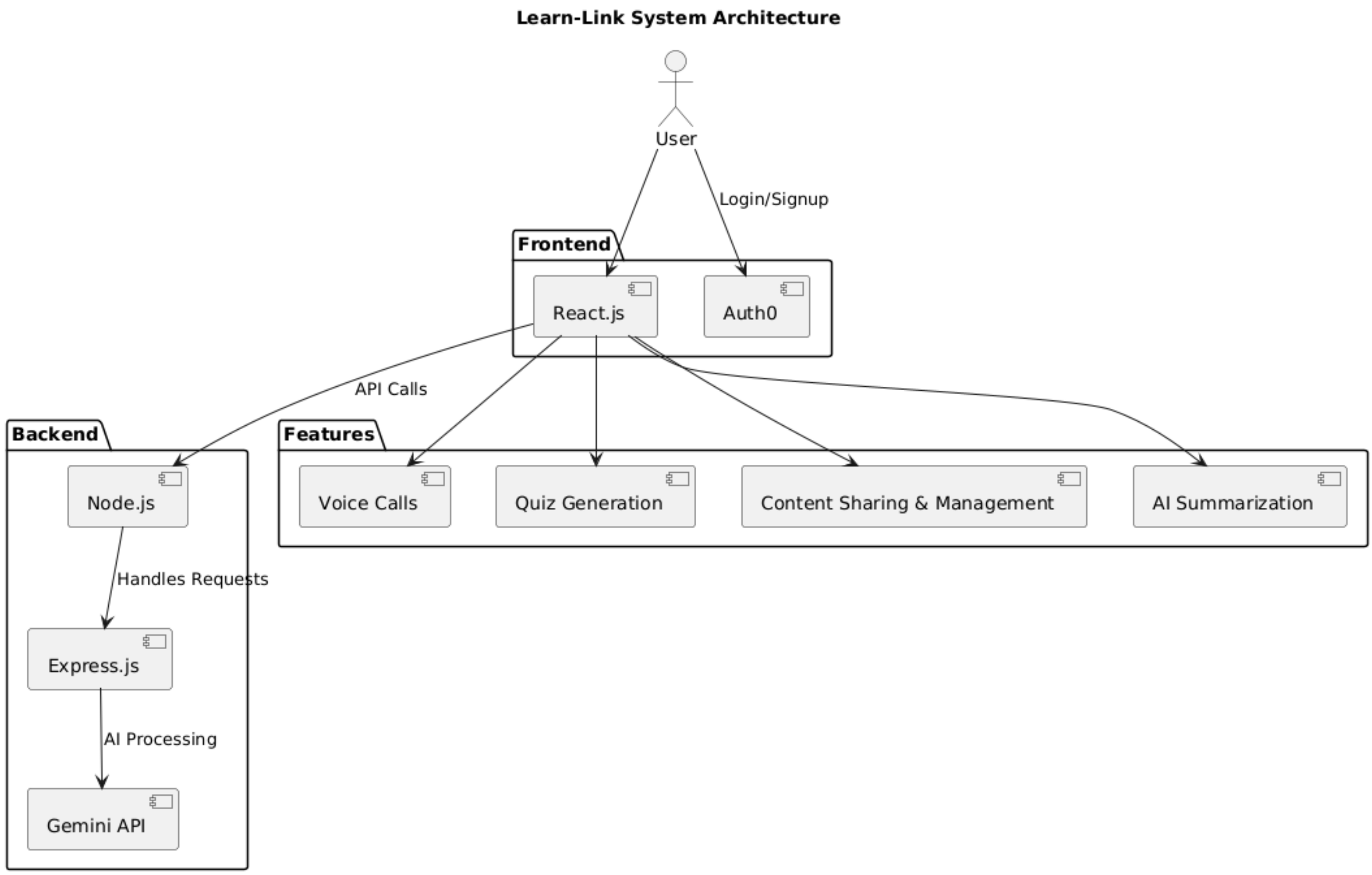

Figure 3.1: Architecture Diagram

3.2 Use case Diagram

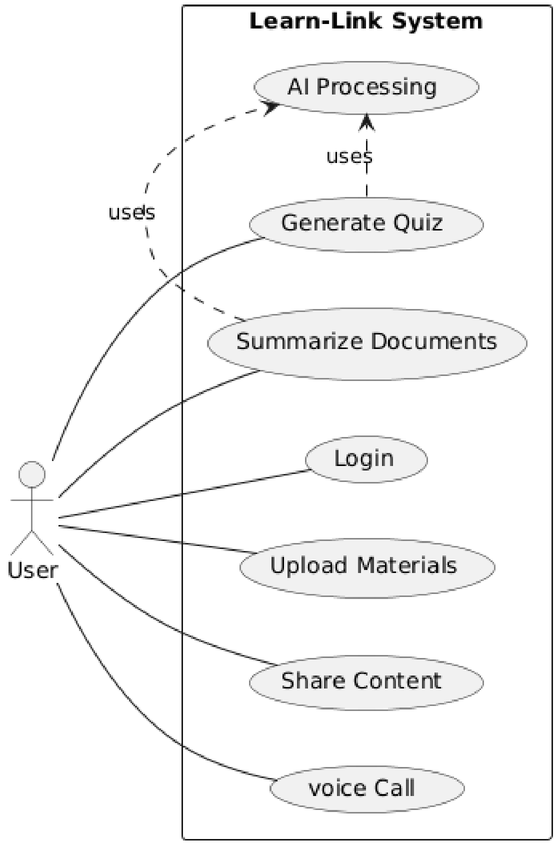

Figure 3.2: Use case diagram

3.3 Data Flow Diagram

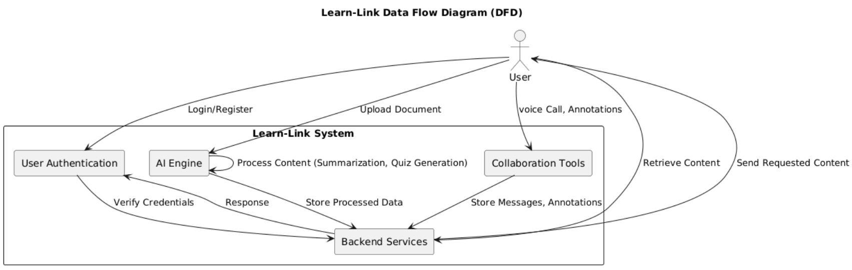

Figure 3.3: Data Flow Diagram

3.4 System Overview

The system architecture of ”Learn-Link” is designed for a seamless and collaborative desktop website experience, utilizing React.js for visual responsiveness and Node.js with Express.js for backend efficiency. It includes real-time communication, AI-powered insights, and structured content management, supporting diverse user classes like students and administrators. Integration of the Gemini API enhances functionality, enabling desktop users to engage in productive discussions and access tailored educational resources.

3.4.1 AI Integration

The Gemini API is an extension for integrating AI-driven tools into Learn-Link's desktop website. To begin, we configure API endpoints in Express.js, ensuring secure HTTPS connections. Next, we integrate the API with React.js components to process user-uploaded content. This setup enables features like document summarization, quiz generation, and focus area extraction, completing the AI integration process for desktop users.

3.4.2 Application Main Interface

To create the main interface for Learn-Link's desktop website, we first designed layouts using React.js optimized for desktop screens. Next, we added interactive elements like chat windows and content upload sections. To enable seamless collaboration, we integrated WebSocket-based communication. Finally, we combined these elements to craft an intuitive and engaging interface for desktop users to navigate.

3.4.3 Graphics And Art Design

In "Learn-Link," the UI assets are carefully designed to enhance the desktop website experience. The process begins with crafting wireframes and mockups, focusing on usability and clarity for larger screens. React.js's component library is used to build desktop-optimized layouts, ensuring consistency across browsers. The design incorporates intuitive navigation, clear labeling, and subtle animations, adding depth and appeal. These elements are tailored for desktop performance, ensuring users enjoy a fluid and visually appealing website that supports efficient knowledge sharing.

3.4.4 Real-Time Collaboration

In "Learn-Link," we utilize Node.js with WebSocket protocols to enable real-time communication on the desktop website, ensuring that live chat, video calls, and co-editing reflect updates instantly. Features like annotations and instant messaging are optimized for desktop responsiveness, allowing dynamic and interactive teamwork among users during learning sessions.

3.4.5 User And Admin Management

A Node.js script manages user authentication and roles via Auth0 in Learn-Link's desktop website. It initializes user sessions with secure tokens. When a user logs in, it verifies credentials and assigns permissions based on their class—normal, registered, or admin. If authentication fails, it restricts access. For admins, it enables content moderation and analytics tracking. After logout, it clears session data and resets access, ensuring security.

Another script handles content management and analytics for desktop users. For registered users, it saves progress and preferences, while admins can generate reports on engagement. The system ensures data privacy with encryption, and updates reflect instantly across the desktop website.

3.4.6 Saving Progress

In "Learn-Link," user progress is displayed in the desktop website interface. When a session ends, completed tasks and quiz scores are shown, providing feedback on achievements. Additionally, personalized recommendations can be accessed from the dashboard. Progress is saved using a distributed SQL database, which stores data securely across servers. This ensures that desktop user preferences and learning history are retained and accessible across sessions.

3.5 System Requirements

3.5.1 Hardware Requirements

The desktop website will run on systems with a minimum of 4 GB RAM and a storage capacity (HDD or SSD) of at least 10 GB. A stable internet connection is necessary for real-time features. Learn-Link has been tested on desktop workstations with these specifications.

3.5.2 Software Requirements

Learn-Link's desktop website supports browsers like Chrome 6+, Safari 5.0.2+, Opera 10.62+, and Internet Explorer 7+, running on Windows 7+, macOS 10.12+, or Ubuntu 16.04+. The backend is deployed via cloud infrastructure using Node.js and Express.js.

3.5.3 React.js

React.js is a powerful JavaScript library for building dynamic, component-based desktop website interfaces. It excels in creating responsive web applications with reusable components, supporting Learn-Link's desktop design. Minimum requirements include a modern desktop browser and Node.js 14+ for development, with 4 GB RAM and 5 GB storage.

3.5.4 Node.js and Express.js

Node.js is a JavaScript runtime for scalable server-side applications, while Express.js simplifies web framework development. They power Learn-Link's desktop website backend, requiring Windows 7+, macOS 10.12+, or Ubuntu 16.04+, with 4 GB RAM, 10 GB storage, and Node.js 14+ installed.

3.5.5 Gemini API

The Gemini API integrates AI-driven tools into Learn-Link's desktop website, enabling summarization and quiz generation. It requires secure API keys managed via Express.js and an active internet connection for cloud-based processing, enhancing the website's educational capabilities.

3.5.6 Auth0

Auth0 provides secure authentication and authorization for Learn-Link's desktop website. It integrates with Node.js to manage user logins and MFA, requiring an internet connection and browser support for HTTPS. Its cloud-based service ensures robust security with minimal local resources.

3.5.7 Visual Studio Code

Visual Studio Code (VS Code) is a versatile code editor by Microsoft, used for developing Learn-Link's desktop website. With extensions for React.js and Node.js, it offers Git integration and debugging. It runs on Windows, macOS, or Linux, needing 4 GB RAM and 5 GB storage.

3.6 Methodology

Using a desktop browser, access the Learn-Link URL to open the website. The main interface will appear, offering options: login, explore, or admin panel. Log in to start a session. Now, engage in collaboration, upload content, and use AI tools. After the session ends, progress will appear, with options to save or exit. The Explore option shows public resources, while the Admin panel manages content and analytics. The Exit option logs you out securely.

Chapter 4

RESULTS

4.1 Case Study: Enhancing Collaborative Learning Using Learn-Link

In the realm of traditional learning, the challenge of creating truly collaborative experiences has long been evident. Students often find themselves isolated in their studies, limited by the constraints of static resources and disconnected platforms. Learn-Link, a desktop website, sought to overcome these barriers by leveraging real-time communication and AI-driven tools.

One of the key defects of traditional learning is the lack of collaboration. Students can feel like solitary learners, rather than active participants in a shared educational journey. Learn-Link addresses this by integrating AI chat, audio calls, and co-editing features, immersing users in a dynamic learning environment. This collaborative setup allows students to feel truly engaged in teamwork, enhancing their overall learning experience.

Another challenge faced by traditional methods is limited interactivity. Many platforms offer only basic tools, leading to repetitive tasks and reduced engagement over time. Learn-Link introduces a variety of interactive elements, from AI-generated quizzes to real-time annotations. This increased interactivity keeps users involved and adds depth to the learning process.

Furthermore, traditional learning often suffers from inefficiency, with outdated resources and fragmented communication. Learn-Link combats this by offering a diverse range of features like document summarization and content tagging. Users can engage in meaningful discussions and manage resources efficiently, ensuring that learning remains fresh and engaging for extended periods.

Learn-Link's innovative approach to education not only overcomes the defects of traditional learning but also offers a wide range of benefits that transcend conventional experiences. One of the most notable advantages of this desktop website is the exceptional level of collaboration it provides. The platform connects users in a shared digital space, allowing them to interact with and explore educational content in ways never before possible. This collaboration can lead to enriched learning outcomes, as users feel more connected to their peers and materials.

Chapter 5

CONCLUSION

Learn-Link stands as a testament to the transformative power of technology in education. By addressing key limitations of traditional learning such as collaboration, interactivity, and efficiency, Learn-Link has set a new standard for productive and engaging learning experiences. Through its innovative use of real-time tools and AI, the desktop website not only provides users with a dynamic and collaborative platform to explore but also offers a glimpse into the future of learning. The success of Learn-Link highlights the potential for technology to revolutionize education, offering a wide range of benefits that transcend traditional learning methods. From enhanced teamwork and personalized insights to streamlined resources and secure access, this platform opens up new possibilities for learners. As web technologies continue to advance, the potential for innovative and transformative learning experiences will only continue to grow, making Learn-Link a pioneering example of the future of education.

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Appendix A

Example Code

A.1 Voice call functionality

```
const socketIo = require("socket.io");

// Setup voice chat functionality
function setupVoiceChat(server) {
  // Store connected users and their status
  const socketsStatus = {};

  // Initialize socket.io with CORS settings
  const io = socketIo(server, {
    cors: {
      origin: [
        "http://localhost:5173", // Local development URL
        "https://learn-link.vercel.app", // Production frontend URL
        process.env.FRONTEND_URL,
        // Additional URL from environment variable (if set)
      ].filter(Boolean), // Remove any undefined/null values
      methods: ["GET", "POST", "OPTIONS"],
      allowedHeaders: ["Content-Type"],
      credentials: true,
    },
  });

  // Create a namespace for voice chat
  const voiceChat = io.of("/voice-chat");
```

```
// Print server information on startup
console.log("[Voice Chat] Server initialized");
console.log("[Voice Chat] Users can connect via websockets");

// Handle socket connections
io.on("connection", function (socket) {
  const socketId = socket.id;
  socketsStatus[socketId] = {
    username: `User ${socketId.substring(0, 5)}`,
    microphone: false,
    mute: false,
    online: true,
  };

  console.log(`[Voice Chat] New connection: ${socketId}`);

  // Handle voice data transmission
  socket.on("voice", function (data) {
    // Process audio data
    let newData = data.split(";");
    newData[0] = "data:audio/ogg;";
    newData = newData[0] + newData[1];

    // Get user info for logging
    const username = socketsStatus[socketId]?.username || "Unknown User";

    // Broadcast to other users who are online and not muted
    let receiverCount = 0;
    for (const id in socketsStatus) {
      if (
        id !== socketId &&
        !socketsStatus[id].mute &&
        socketsStatus[id].online
      ) {
        socket.broadcast.to(id).emit("send", newData);
```

```
          receiverCount++;
        }
      }

      // Log transmission (but not too frequently)
      if (Math.random() < 0.01) {
        // Log only 1% of transmissions to avoid spam
        console.log(
          `[Voice Chat] ${username} speaking to ${receiverCount} users`
        );
      }
    });

    // Handle user status updates
    socket.on("userInformation", function (data) {
      // Store previous state to detect changes
      const previousState = { ...(socketsStatus[socketId] || {}) };

      // Update user information
      socketsStatus[socketId] = {
        ...data,
        lastUpdated: new Date().toISOString(),
      };

      // Log status changes
      if (previousState.username !== data.username) {
        console.log(
          `[Voice Chat] User renamed: ${
            previousState.username || "New user"
          } → ${data.username}`
        );
      }

      if (previousState.online !== data.online) {
        console.log(
          `[Voice Chat] User ${data.username} is now ${
```

```
            data.online ? "online" : "offline"
          }`
        );
      }

      if (previousState.microphone !== data.microphone) {
        console.log(
          `[Voice Chat] User ${data.username} microphone is now ${
            data.microphone ? "on" : "off"
          }`
        );
      }

      // Send updated user list to all clients
      io.sockets.emit("usersUpdate", socketsStatus);
    });

    // Handle disconnection
    socket.on("disconnect", function () {
      const username = socketsStatus[socketId]?.username || "Unknown User";
      console.log(`[Voice Chat] Disconnected: ${username} (${socketId})`);

      delete socketsStatus[socketId];
      io.sockets.emit("usersUpdate", socketsStatus);
    });
  });

  return io;
}

module.exports = setupVoiceChat;
```

A.2 AI service functions

```
require('dotenv').config();
const { GoogleGenerativeAI } = require("@google/generative-ai");
const { GoogleAIFileManager } = require("@google/generative-ai/server");
const fs = require('fs').promises;

if (!process.env.GEMINI_API_KEY) {
  throw new Error('GEMINI_API_KEY is not set in environment variables');
}

const genAI = new GoogleGenerativeAI(process.env.GEMINI_API_KEY);
const fileManager = new GoogleAIFileManager(process.env.GEMINI_API_KEY);
const model = genAI.getGenerativeModel({ model: "models/gemini-2.0-flash" });

async function generateAIContent(prompt) {
  try {
    const result = await model.generateContent(prompt);
    return result.response.text();
  } catch (error) {
    console.error('Error generating AI content:', error);
    throw error;
  }
}

async
function processPdfContent(pdfBuffer, prompt = 'Summarize this document') {
  try {
    const result = await model.generateContent([
      {
        inlineData: {
          data: pdfBuffer.toString('base64'),
          mimeType: "application/pdf",
        }
      },
      prompt
```

```
    ]);
    return result.response.text();
  } catch (error) {
    console.error('Error processing PDF:', error);
    throw error;
  }
}

async function localPdfToPart(buffer, displayName) {
  const tempPath = `/tmp/${displayName}-${Date.now()}.pdf`;
  try {
    await fs.writeFile(tempPath, buffer);
  } catch (error) {
    console.error('Error writing temporary PDF file:', error);
    throw error;
  }
  try {
    const uploadResult = await fileManager.uploadFile(
      tempPath,
      {
        mimeType: "application/pdf",
        displayName,
      },
    );

    return {
      fileData: {
        fileUri: uploadResult.file.uri,
        mimeType: uploadResult.file.mimeType,
      },
    };
  } finally {
    // Cleanup temp file
    await fs.unlink(tempPath).catch(console.error);
  }
}
```

```
async function comparePdfs(pdf1Buffer, pdf2Buffer) {
  try {
    const result = await model.generateContent([
      await localPdfToPart(pdf1Buffer, 'PDF 1'),
      await localPdfToPart(pdf2Buffer, 'PDF 2'),
      'What are some important topics in these documents?,
      generate a focus area for exams as points',
    ]);
    return result.response.text();
  } catch (error) {
    console.error('Error comparing PDFs:', error);
    throw error;
  }
}

/**
 * Generates quiz questions based on PDF content
 * @param {Array} pdfFiles - Array of PDF objects with buffer and filename
 * @returns {Object} Quiz object with questions array
 */
async function generateQuiz(pdfFiles) {
  try {
    console.log(`[generateQuiz] Processing ${pdfFiles.length} PDF files`);

    // Generate questions for each PDF
    const allQuestions = [];

    for (let i = 0; i < pdfFiles.length; i++) {
      // Check if we're receiving a proper object with buffer and filename
      let buffer, filename;

      if (pdfFiles[i].buffer && pdfFiles[i].filename) {
        // If properly structured object
        buffer = pdfFiles[i].buffer;
        filename = pdfFiles[i].filename;
```

```
} else {
  // If just a buffer
  buffer = pdfFiles[i];
  filename = `Document ${i+1}`;
}

console.log(`[generateQuiz] Processing file ${i+1}: ${filename}`);

const prompt = `
  You are an expert teacher creating a quiz based on this PDF document.

  Create 5 multiple-choice questions that test understanding
  of key concepts in the document.
  Each question must:
  1. Be based directly on specific content in this document
  2. Have four answer options (A, B, C, D)
  3. Have one clearly correct answer
  4. Include a brief explanation of the correct answer

  Format your response as a valid JSON array with this structure:
  [
    {
      "question": "Question text here?",
      "options": ["First option", "Second option",
      "Third option", "Fourth option"],
      "correctAnswer": 0,
      "explanation": "Why this is the correct answer,
      with reference to document content"
    }
  ]

  Remember: Create exactly 5 questions, each with 4 options.
  Make sure the content is specifically from this document.
  Only provide the JSON array, no other text.
`;
```

```
try {
  const result = await model.generateContent([
    {
      inlineData: {
        data: buffer.toString('base64'),
        mimeType: "application/pdf",
      }
    },
    prompt
  ]);

  const responseText = result.response.text();
  console.log('[generateQuiz] Received AI response for file:', filename);

  // Extract JSON if it's wrapped in markdown code blocks
  const jsonMatch = responseText.match
  (/```(?:json)?\s*([\s\S]*?)\s*```/) || [null, responseText];
  const jsonStr = jsonMatch[1].trim();

  try {
    const parsedQuestions = JSON.parse(jsonStr);
    console.log(`[generateQuiz] Successfully parsed
    ${parsedQuestions.length} questions for ${filename}`);

    // Display a sample question in console for debugging
    console.log(`[generateQuiz] Sample question:`, parsedQuestions[0]);

    // Add file identifier to each question
    const questionsWithSource = parsedQuestions.map(q => ({
      ...q,
      source: filename
    }));

    allQuestions.push(...questionsWithSource);
  } catch (parseError) {
    console.error('[generateQuiz] Failed to parse JSON:', parseError);
```

```
        console.log('[generateQuiz] Raw response:', responseText);

        // Create fallback questions for this document
        const fallbackQuestions = [
          {
            question: `What is the main topic of ${filename}?`,
            options: [
              "The primary subject of the document",
              "An unrelated concept",
              "A minor detail",
              "None of the above"
            ],
            correctAnswer: 0,
            explanation: "This is a fallback question.",
            source: filename
          }
        ];

        allQuestions.push(...fallbackQuestions);
      }
    } catch (error) {
      console.error(`[generateQuiz] Error generating questions:`, error);

      // Add single fallback question
      allQuestions.push({
        question: `What information can be found in ${filename}?`,
        options: ["Main content", "Unrelated information", "No specific content",
        "Random data"],
        correctAnswer: 0,
        explanation: "This is a fallback question due to an error.",
        source: filename
      });
    }
  }

  console.log(`[generateQuiz] Total questions generated: ${allQuestions.length}`);
```

```
// Ensure we have at least 5 questions total
if (allQuestions.length < 5) {
  const moreNeeded = 5 - allQuestions.length;
  console.log(`[generateQuiz] Adding ${moreNeeded} generic questions
  to reach minimum of 5`);

  for (let i = 0; i < moreNeeded; i++) {
    allQuestions.push({
      question: `Generic Question ${i+1}: What approach is best for
      studying the content?`,
      options: ["Understanding key concepts", "Memorizing random facts",
      "Ignoring main topics", "Not reading the document"],
      correctAnswer: 0,
      explanation: "Understanding key concepts is always the best approach.",
      source: "General"
    });
  }
}

// Print all generated questions to terminal for debugging
console.log('\n=== GENERATED QUIZ QUESTIONS ===');
allQuestions.slice(0, 5).forEach((q, i) => {
  console.log(`\nQ${i+1} [${q.source}]: ${q.question}`);
  q.options.forEach((opt, j) => {
    const marker = j === q.correctAnswer ? '' : ' ';
    console.log(`  ${marker} ${opt}`);
  });
});
console.log('\n================================\n');
```

```
    // Return only 5 questions
    return { questions: allQuestions.slice(0, 5) };

  } catch (error) {
    console.error('[generateQuiz] Error in quiz generation:', error);
    throw error;
  }
}

module.exports = {
  generateAIContent,
  processPdfContent,
  comparePdfs,
  generateQuiz
};
```

Appendix B

Screenshots

Figure B.1: Title Page

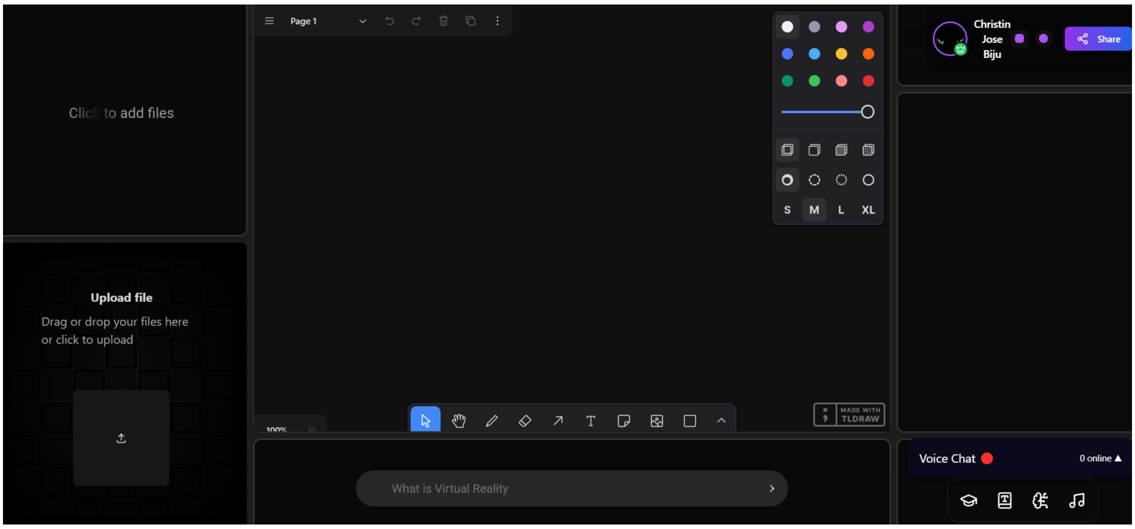

Figure B.2: Dashboard

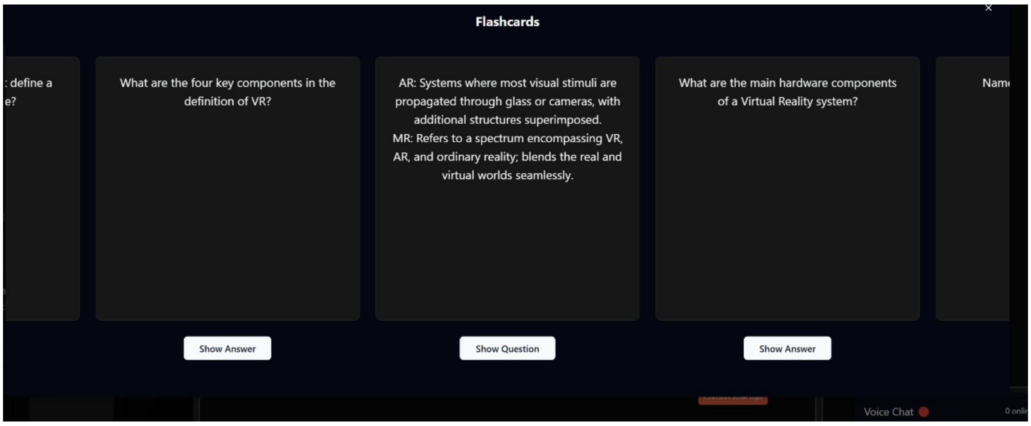

Figure B.3: Flashcard

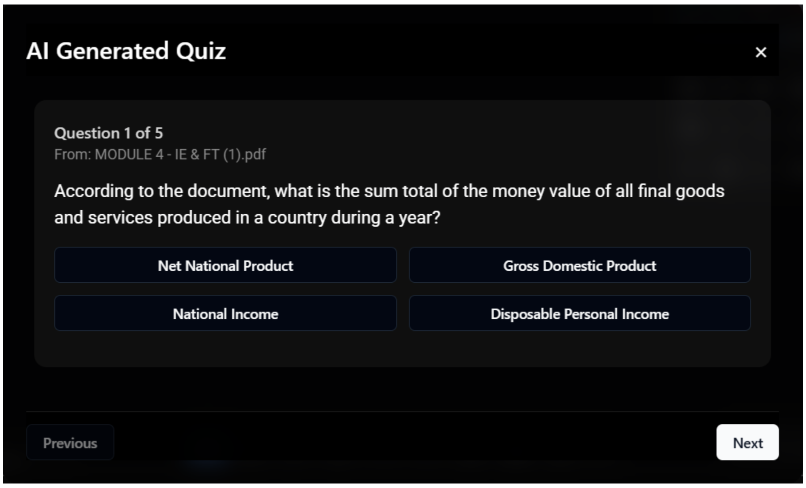

Figure B.4: Ai quiz

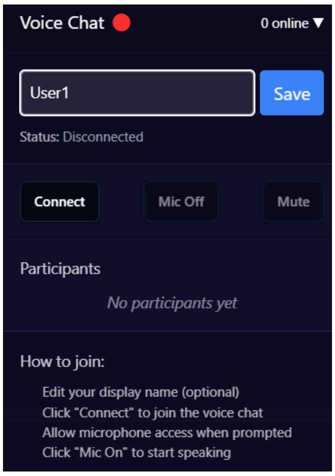

Figure B.5: voice call